

INTRODUCTION TO KAFKA - NOTES

KAFKA

Monday, June 19, 2023
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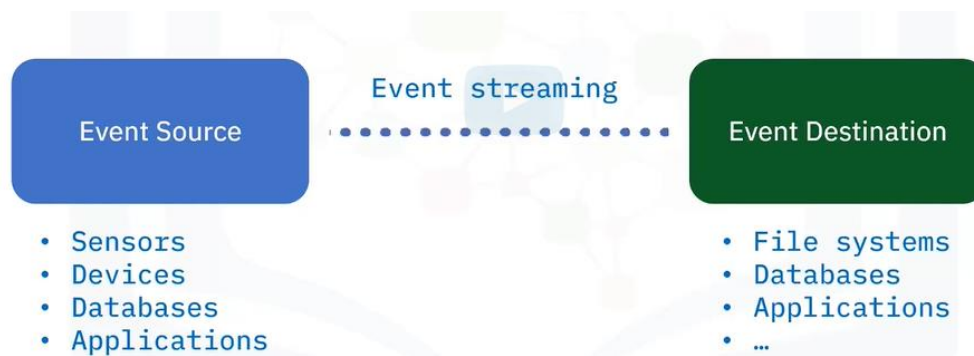
EVERYTHING YOU NEED TO KNOW ABOUT KAFKA:

Apache Kafka is a very popular open-source event streaming pipeline.

Kafka Streams API is a client library supporting you with data processing in event streaming pipelines.

Event: Event is an entity with observable updates over time. Ex. Coordinates of moving car, wind speed at a place etc

Event streaming: Continuous flow of events in near real time from source to destination.

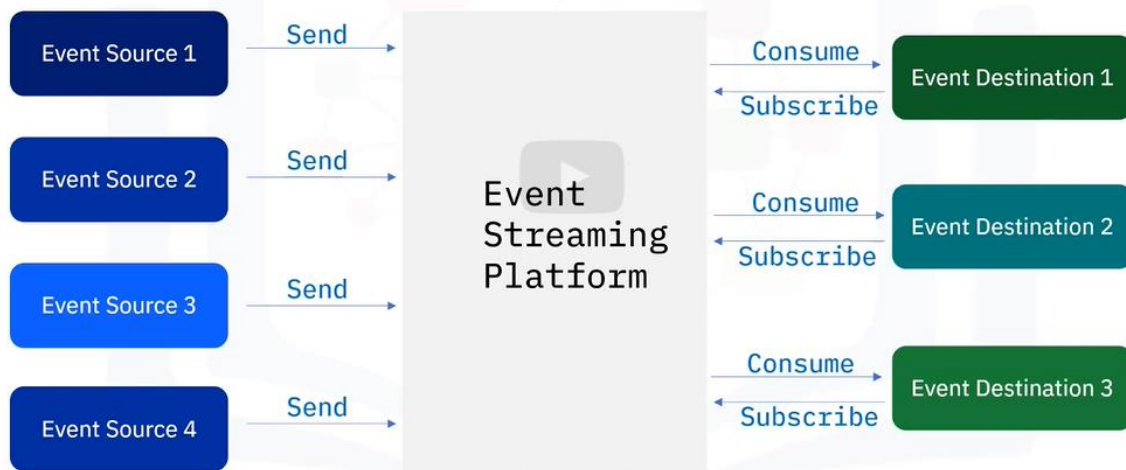


Event Stream platform:

==> ESP acts as a middle man to handle the source & destination for the real time events that were generated by providing a platform.

==>Destination: It can subscribe to a particular event source so that it can consume data only from the source.

Event Streaming Platform (ESP)

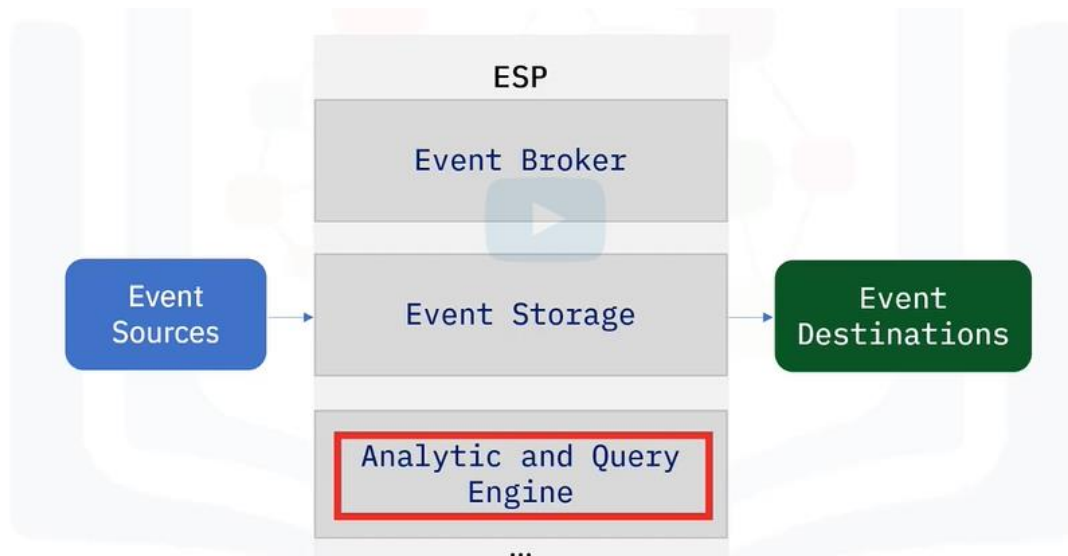


Components of ESP:

==>Event broker: Used to receive events from the event source

Event storage: Used to store the received events which can later be consumed by the destination

Analytics & query engine: used to analyze/query the stored events

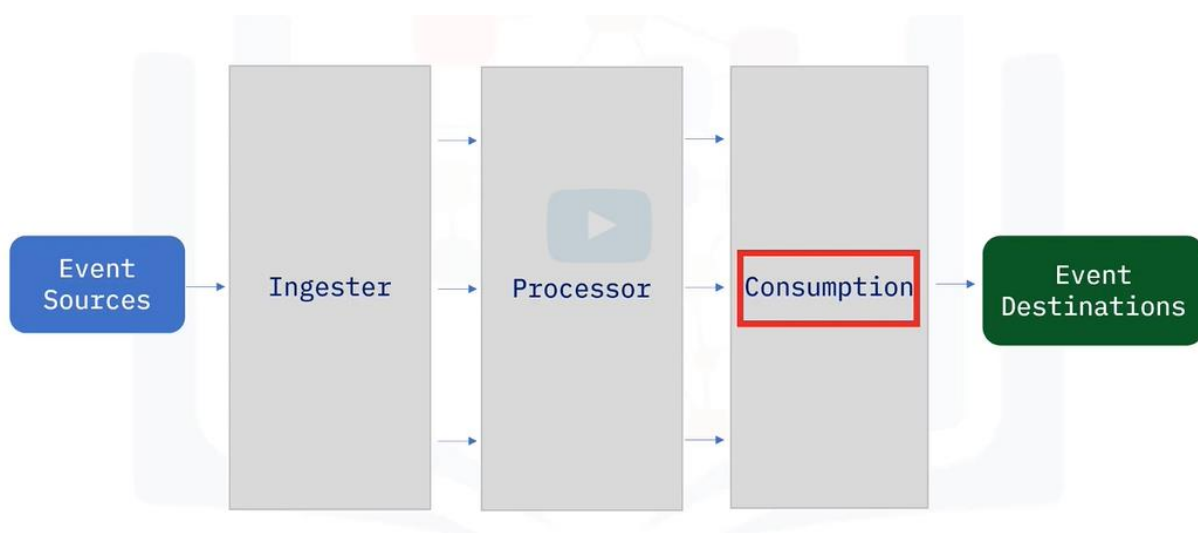


Event Broker:

Ingestor: Ingests the events from various sources

Processes: Processes the ingested data like serializing/de-serializing, compression/de-compression, encryption/de-encryption

Consumption: efficiently distributes the events consumed to their correct destinations.



Popular ESP:



Kafka Architecture:

Distributed servers: They serve the purpose of receiving and storing the events from the sources.

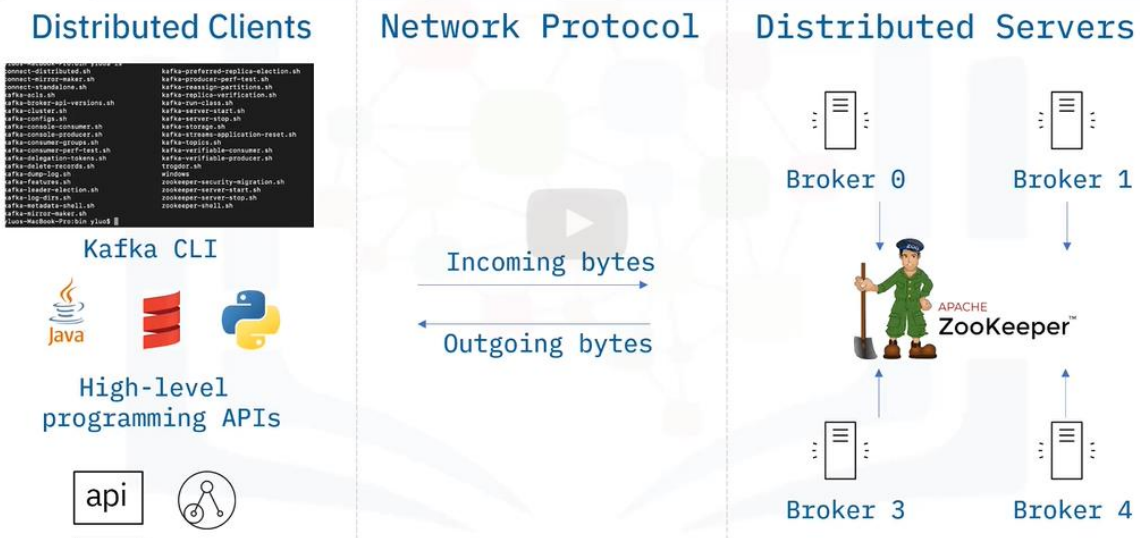
Zoo keeper: To manage all the brokers and ensures they work in efficient way.

=> ZooKeeper is responsible for the overall management of Kafka cluster. It monitors the Kafka brokers and notifies Kafka if any broker or partition goes down, or if a new broker or partition goes up.

Network protocol: TCP (transmission control protocol) is used to facilitate sending & receiving of events.

Distributed clients: Facilitates to communicate with the brokers.

Kafka architecture



Features of kafka:

- ==>Distributed system: It can process parallelly and works in distributed manner.
- ==>Highly scalable : It can handle huge amount of data by scaling up on the go.
- ==>Highly reliable: Stores the data in partition and replicates the data to be fault tolerant.
- ==>Permanent persistency: It will store the data permanently (long time) and can be consumed whenever destination is ready.
- ==>Open source: it is a open source software and can be customized to ur needs.

- Distribution system
- Highly scalable
- Highly reliable
- Permanent persistency
- Open source



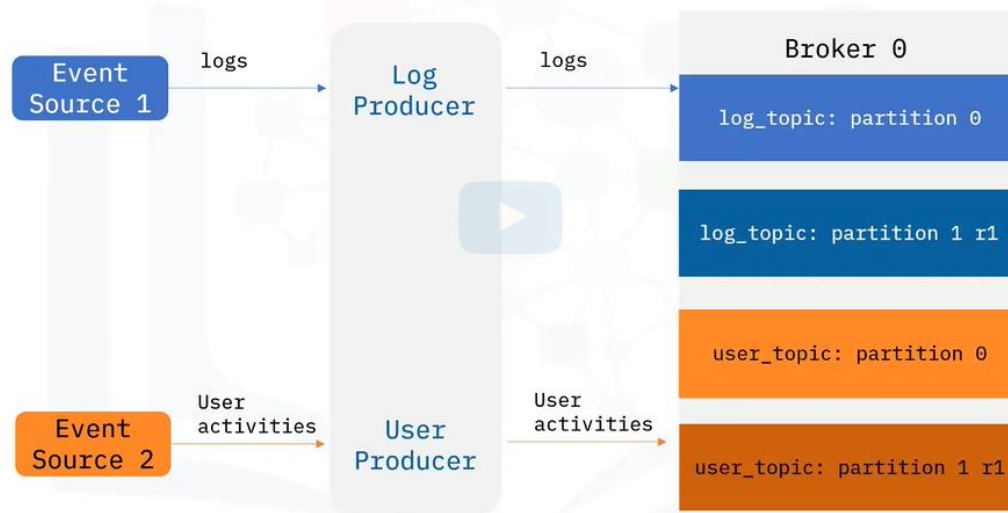
Topic: Database to store specific events like log topic, gps topic, sensor topic etc

Kafka producer

- Client applications that publish events to topic partition
- An event can be optionally associated with a key
- Events associated with the same key will be published to the same topic partition
- Events not associated with any key will be published to topic partitions in rotation

Kafka producer:

1. Events will be published to specific event topic.
2. The events will be partitioned and replicated to be spread across different brokers
3. If one of the broker is down then events will be sent to the replicated event topics.
4. Its good to associate every topic with a key

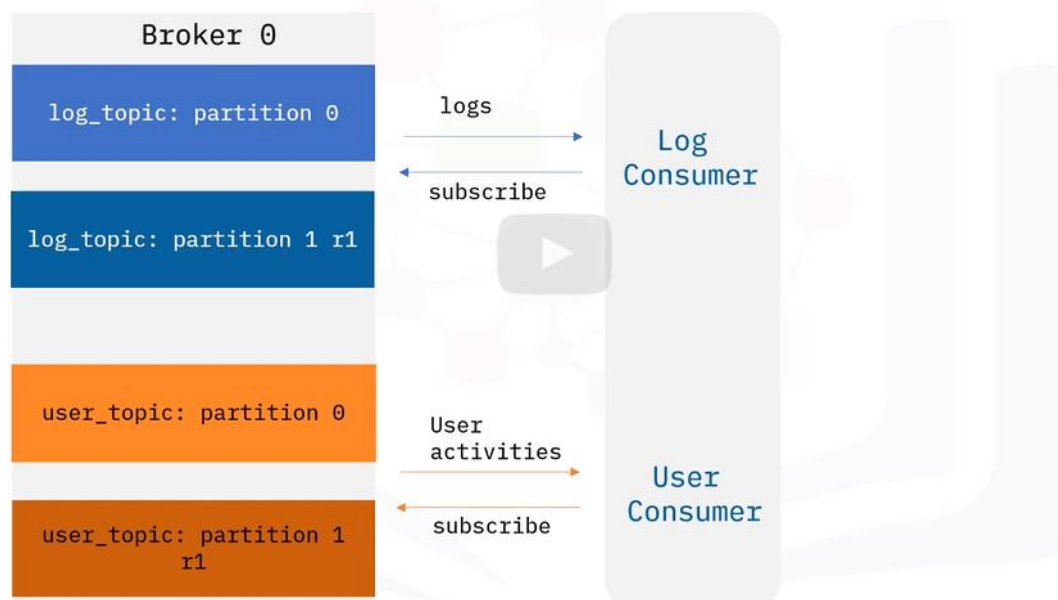


Kafka consumer: It consumes the events from the kafka producer.

=>Data will be read by the kafka consumer based on the events it is subscribed to

Kafka consumer

- Consumers are clients subscribed to topics
- Consume data in the same order
- Store an offset record for each partition
- Offset can be reset to zero to read all events from the beginning again



Kafka Streams API: Used to facilitate data processing in events streaming .

--> Used to process the data stored in kafka topics.

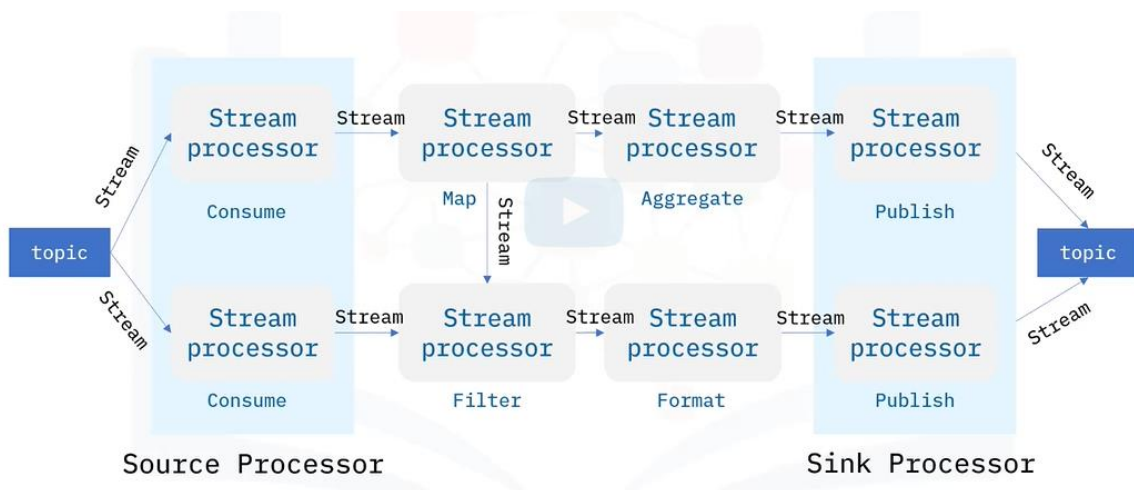
-->It can process only one record at a time

Stream processing flow:

Source processor: It acts like a consumer where it consumes the events from the topic.

Processing: Map, filter, aggregate, format (list of processes that can transform the data received from source)

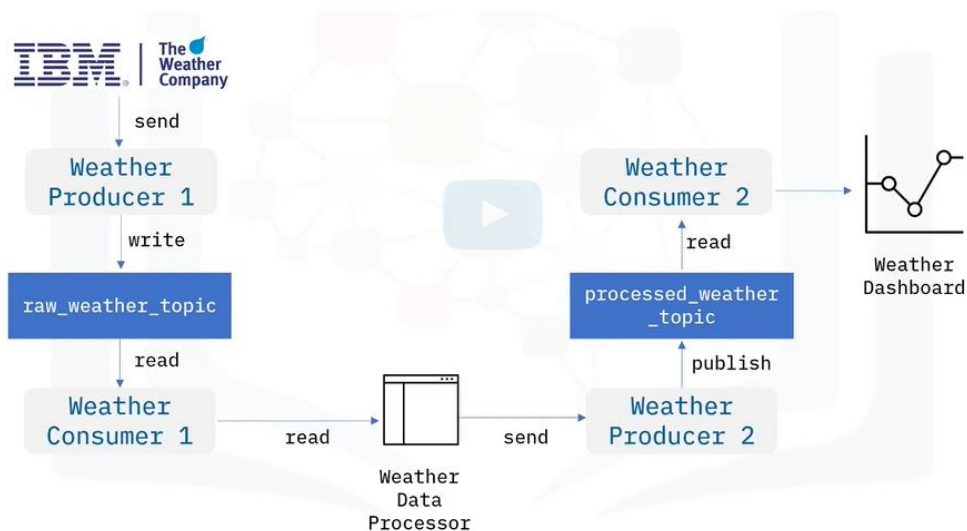
Sink processor: It acts like a producer and sends the final processed data to destination topic



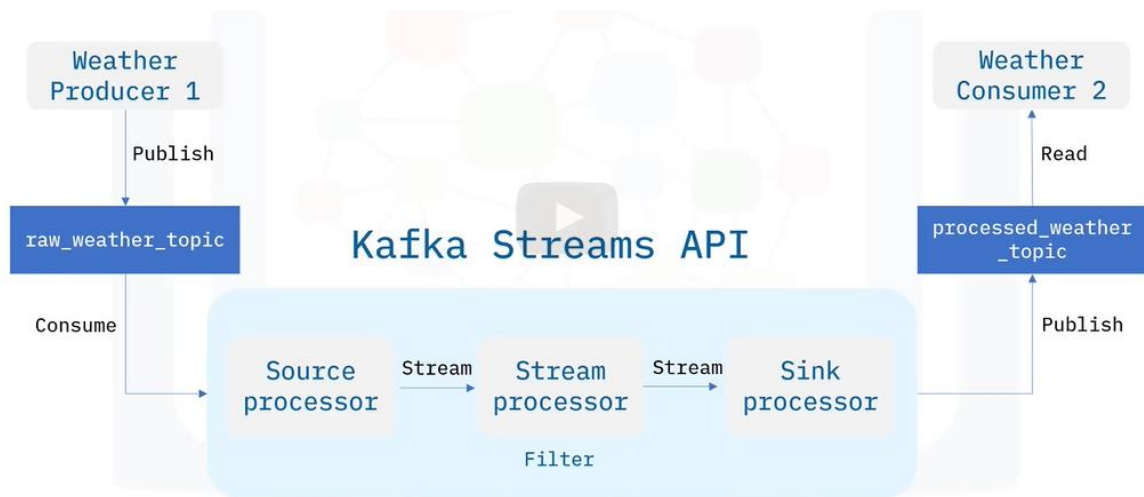
=> with the help of kafka streaming API it is easy to even process the data and store it in a desired format that suits our need.

Simple weather streaming project:

Project Flow diagram:



Using kafka streaming API to filter the highest temperature based on region on a given date time:



Kafka CLI options:

Options for kafka features:

Option	Description
-----	-----
--bootstrap-server <String: server to connect to>	REQUIRED: A comma-separated list of host:port pairs to use for establishing the connection to the Kafka cluster.
--command-config [String: command config property file]	Property file containing configs to be passed to Admin Client. This is used with --bootstrap-server option when required.
--describe	Describe supported and finalized features from a random broker.
--downgrade-all	Downgrades all finalized features to the maximum version levels known to the tool. This command deletes unknown features from the list of finalized features in the cluster, but it is guaranteed to not add a new feature.
--dry-run	Performs a dry-run of upgrade/downgrade mutations to finalized feature without applying them.
--help	Print usage information.
--upgrade-all	Upgrades all finalized features to the maximum version levels known to the tool. This command finalizes new features known to the tool that were never finalized previously in the

cluster, but it is guaranteed to not delete any existing feature.

--version Display Kafka version.

Options for kafka producer:

Option	Description
-----	-----
--batch-size <Integer: size>	Number of messages to send in a single batch if they are not being sent synchronously. (default: 200)
--bootstrap-server <String: server to connect to>	REQUIRED unless --broker-list (deprecated) is specified. The server (s) to connect to. The broker list string in the form HOST1:PORT1,HOST2:PORT2.
--broker-list <String: broker-list>	DEPRECATED, use --bootstrap-server instead; ignored if --bootstrap-server is specified. The broker list string in the form HOST1:PORT1, HOST2:PORT2.
--compression-codec [String: compression-codec]	The compression codec: either 'none', 'gzip', 'snappy', 'lz4', or 'zstd'. If specified without value, then it defaults to 'gzip'
--help	Print usage information.
--line-reader <String: reader_class>	The class name of the class to use for reading lines from standard in. By default each line is read as a separate message. (default: kafka.tools.ConsoleProducer\$LineMessageReader)
--max-block-ms <Long: max block on send>	The max time that the producer will block for during a send request (default: 60000)
--max-memory-bytes <Long: total memory in bytes>	The total memory used by the producer to buffer records waiting to be sent to the server. (default: 33554432)
--max-partition-memory-bytes <Long: memory in bytes per partition>	The buffer size allocated for a partition. When records are received which are smaller than this size the producer will attempt to optimistically group them together until this size is reached. (default: 16384)
--message-send-max-retries <Integer>	Brokers can fail receiving the message for multiple reasons, and being

unavailable transiently is just one of them. This property specifies the number of retries before the producer give up and drop this message. (default: 3)

--metadata-expiry-ms <Long: metadata expiration interval> The period of time in milliseconds after which we force a refresh of metadata even if we haven't seen any leadership changes. (default: 300000)

--producer-property <String: producer_prop> A mechanism to pass user-defined properties in the form key=value to the producer.

--producer.config <String: config file> Producer config properties file. Note that [producer-property] takes precedence over this config.

--property <String: prop> A mechanism to pass user-defined properties in the form key=value to the message reader. This allows custom configuration for a user-defined message reader. Default properties include:
 parse.key=true|false
 key.separator=<key.separator>
 ignore.error=true|false

--request-required-acks <String: request required acks> The required acks of the producer requests (default: 1)

--request-timeout-ms <Integer: request timeout ms> The ack timeout of the producer requests. Value must be non-negative and non-zero (default: 1500)

--retry-backoff-ms <Integer> Before each retry, the producer refreshes the metadata of relevant topics. Since leader election takes a bit of time, this property specifies the amount of time that the producer waits before refreshing the metadata. (default: 100)

--socket-buffer-size <Integer: size> The size of the tcp RECV size. (default: 102400)

--sync If set message send requests to the brokers are synchronously, one at a time as they arrive.

--timeout <Integer: timeout_ms> If set and the producer is running in asynchronous mode, this gives the maximum amount of time a message will queue awaiting sufficient batch size. The value is given in ms. (default: 1000)

--topic <String: topic> REQUIRED: The topic id to produce

messages to.
--version Display Kafka version.

Options for kafka consumer:

Option	Description
-----	-----
--bootstrap-server <String: server to connect to>	REQUIRED: The server(s) to connect to.
--consumer-property <String: consumer_prop>	A mechanism to pass user-defined properties in the form key=value to the consumer.
--consumer.config <String: config file>	Consumer config properties file. Note that [consumer-property] takes precedence over this config.
--enable-systest-events	Log lifecycle events of the consumer in addition to logging consumed messages. (This is specific for system tests.)
--formatter <String: class>	The name of a class to use for formatting kafka messages for display. (default: kafka.tools.DefaultMessageFormatter)
--from-beginning	If the consumer does not already have an established offset to consume from, start with the earliest message present in the log rather than the latest message.
--group <String: consumer group id>	The consumer group id of the consumer.
--help	Print usage information.
--isolation-level <String>	Set to read_committed in order to filter out transactional messages which are not committed. Set to read_uncommitted to read all messages. (default: read_uncommitted)
--key-deserializer <String: deserializer for key>	
--max-messages <Integer: num_messages>	The maximum number of messages to consume before exiting. If not set, consumption is continual.
--offset <String: consume offset>	The offset id to consume from (a non-negative number), or 'earliest' which means from beginning, or 'latest' which means from end (default: latest)
--partition <Integer: partition>	The partition to consume from. Consumption starts from the end of

the partition unless '--offset' is specified.

--property <String: prop> The properties to initialize the message formatter. Default properties include:
print.timestamp=true|false
print.key=true|false
print.offset=true|false
print.partition=true|false
print.headers=true|false
print.value=true|false
key.separator=<key.separator>
line.separator=<line.separator>
headers.separator=<line.separator>
null.literal=<null.literal>
key.deserializer=<key.deserializer>
value.deserializer=<value.deserializer>
header.deserializer=<header.deserializer>

Users can also pass in customized properties for their formatter; more specifically, users can pass in properties keyed with 'key.deserializer.', 'value.deserializer.' and 'headers.deserializer.' prefixes to configure their deserializers.

--skip-message-on-error If there is an error when processing a message, skip it instead of halt.
--timeout-ms <Integer: timeout_ms> If specified, exit if no message is available for consumption for the specified interval.
--topic <String: topic> The topic id to consume on.
--value-deserializer <String: deserializer for values>
--version Display Kafka version.
--whitelist <String: whitelist> Regular expression specifying whitelist of topics to include for consumption.

COMMANDS TO FOLLOW:

==> All the shell scripts are available in the cloud lab. You can go through by

1.cd /home/project/kafka_2.12-2.8.0/bin

2.ls

3.you can view the list of shell scripts available along with the various options available within them

CREATE A TOPIC:

bin/kafka-topics.sh --create --topic news --bootstrap-server localhost:9092 => creates a topic named "news" which runs on the bootstrap server which runs on the localhost.

CREATE PRODUCER

bin/kafka-console-producer.sh --topic news --bootstrap-server localhost:9092

CREATE CONSUMER:

bin/kafka-console-consumer.sh --topic news --from-beginning --bootstrap-server localhost:9092

In the cloud labs, you will be able to create a topic and start a producer where you can send the messages as events
Then start a consumer in different terminal where you can receive the messages sent from the producer

Producer:

```
theia@theiadocker-vaishnavic14:/home/project$ cd kafka_2.12-2.8.0
theia@theiadocker-vaishnavic14:/home/project/kafka_2.12-2.8.0$ bin/kafka-console-producer.sh
--topic weather --from-beginning --bootstrap-server localhost:9092
this is weather reporting from abnglore
todays temperature is 28 degrees
it is veru much better than previous month
```

Consumer:

```
theia@theiadocker-vaishnavic14:/home/project$ cd kafka_2.12-2.8.0
theia@theiadocker-vaishnavic14:/home/project/kafka_2.12-2.8.0$ bin/kafka-console-consumer.sh --topic wea
ther --from-beginning --bootstrap-server localhost:9092
this is weather reporting from abnglore
todays temperature is 28 degrees
it is veru much better than previous month
□
```